Neural networks trained on compact group composition learn the global form of the group

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Abstract

Small neural networks trained to identify products in a compact Lie group learn the irreducible representations of the group. Across ten group forms drawn from $\mathrm{SU}(2)$, $\mathrm{SU}(3)$, $\mathrm{Spin}(4)$, and $\mathrm{Spin}(5)$ and their central quotients, with each element presented through a fixed generic encoding, the trained features concentrate on the matrix coefficients of a small faithful representation of the task group, the same one in every seed, and half-integer spins appear precisely when the encoding distinguishes the two sheets of the double cover. The learned features therefore determine the isomorphism type of the group and not merely its Lie algebra. Ablations show that each isotypic component is necessary for exactly the center discriminations its central character supports. A test based on the two-sided translation action shows that the features span the full matrix-coefficient space of the representation, whose real dimension the real Schur lemma prescribes, and that they admit commuting linear left and right translation actions. A frame of two columns of a rotation matrix determines the rotation with fewer coordinates and admits a linear left action, yet in matched training on $\mathrm{SO}(3)$ the 9-dimensional coefficient space attains a larger margin than the 6-dimensional frame at 17% lower total parameter norm, so regularized training selects the coefficient space as the cheaper complete mechanism. Under regression onto heat-kernel targets, the same architecture fills isotypic components in order of Casimir eigenvalue up to a truncation point that moves with the weight decay. One finding is negative. On $\mathrm{SO}(5)$, a stable vector-plus-adjoint mixture survives every replication, so the minimal-representation description fails as stated in one of the ten group forms.

1 Introduction

By the Peter–Weyl theorem, the matrix coefficients of the irreducible unitary representations of a compact Lie group span a dense subspace of L^2 of the group [Bröcker and tom Dieck, 1985, Ch. III]. For *finite* groups, mechanistic interpretability has established that small networks trained on the multiplication table represent elements through matrix coefficients of a few irreducible representations [Chughtai et al., 2023, Sec. 4], that competing descriptions of the same models exist [Stander et al., 2024, Sec. 5], and that per-argument equivariance reconciles them Wu et al. [2025]. All of this work presupposes a finite table and one learned embedding per element, the setting of the grokking literature Power et al. [2022], Nanda et al. [2023], Gromov [2023], Zhong et al. [2023], the associated theory proves emergence of irreducible features for finite groups Morwani et al. [2024], Marchetti et al. [2024], and equivariant architectures build the harmonics into the network Cohen and Welling [2016], Geiger and Smidt [2022]. Whether this representation theory emerges in a generic network when the group is a continuum has remained open.

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In this paper, we answer that question and report the regularities that the analysis uncovered along the way. We train encoder-factorized scorers to identify products in compact groups, with each element presented through a fixed generic encoding, and we analyze the learned features by isotypic decomposition, ablation, activation patching, and a test built on the two-sided translation action of the group on its own matrix coefficients. Four findings result, and the last revises the first.

The learned representation determines the global form of the group. For each central subgroup $Z \subseteq Z(G)$, averaging the encoding over Z realizes the same task on the quotient G/Z, so the isomorphism type of the group becomes an experimental variable while the Lie algebra stays fixed. Across ten group forms (Table 1), the trained features concentrate 93–98% of their variance on the matrix coefficients of a small faithful representation of the task group, the same one in every seed, and every isotypic component that the descent rule forbids stays at the measurement floor. The selection is causal. Ablating the learned representation collapses ranking to chance and removes exactly the center discriminations its central character supports, while equal-rank ablations elsewhere leave the model intact (Section 3).

The learned features span the full matrix-coefficient space, and cost selects it. The active subspace of the features has exactly the real dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} \mathrm{End}_G(W)$ of the full coefficient space of the learned representation W, and fitted left and right translation operators reach held-out $R^2 \geq 0.98$ and commute, which characterizes the full coefficient space inside its isotypic component (Lemma 2). A frame of two columns determines a rotation with six coordinates and admits an exactly linear left action, and its right action fails to be linear. In matched training on $\mathrm{SO}(3)$, the 9-dimensional coefficient space attains a larger margin than the 6-dimensional frame at 17% lower parameter norm (Section 4). Regularized training selects the coefficient space because composition and scoring become bilinear on it, and the saving on those maps exceeds the cost of the extra coordinates.

Regression targets reveal a spectral bias ordered by the Casimir eigenvalue. Trained to regress heat-kernel values $K_t(ab\,c^{-1})$, the same architecture recovers the required isotypic components at their required weights up to a sharp truncation point, the truncation advances in order of Casimir eigenvalue during training, in analogy with the spectral bias of Rahaman et al. [2019], and the truncation moves with the regularization strength. On a band-limited target, weight decay 1.0 prunes an in-support component that weight decay 0.1 retains (Section 5).

One finding is negative. On $\mathrm{SO}(5)$, a stable vector-plus-adjoint mixture survives five seeds and every conditioning variation we ran, so the minimal-representation description fails as stated in one of ten group forms. We report the failure in Section 6 together with the two hypotheses that remain open.

Our analysis methods are standard. The new ingredients are the object, a continuum group with no embedding table, the quotient construction that turns the isomorphism type into an experimental variable, and the two-sided translation test of Section 4. The rest of this paper is organized as follows. In Section 2, we fix the task and the analysis methods. In Sections 3 to 5, we present the three positive findings. In Section 6, we present the negative finding and a selection conjecture. Background from representation theory and harmonic analysis is collected in Appendix A, proofs are in Appendix B, experimental details in Appendix C, and complete result tables and additional figures in Appendix D.

2 Task, encodings, and analysis methods

Throughout, G denotes a compact Lie group with Haar probability measure, $Z \subseteq Z(G)$ a central subgroup, W an irreducible real representation of G with commutant $\mathrm{End}_G(W) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, $z(g) \in \mathbb{R}^{128}$ the trained argument-encoder activations, and χ_ρ the character of an irreducible unitary representation ρ.

Task. Given encodings of $a, b \in G$ and of $K = 64$ candidates, the model must identify the product. The positive candidate is $(ab)\exp(\xi)$ with ξ a centered Gaussian of scale $\sigma = 0.2$ on the Lie algebra, the $K - 1$ negatives are Haar samples excluded from a ball around the positive class, and the loss is cross-entropy over the K scores. The Bayes-optimal rule ranks candidates by distance to ab and attains accuracy 0.951 under the $\mathrm{SU}(2)$ sampling. Training pairs come from a pool of 2048 Haar elements, and we report ranking accuracy on pairs of fresh elements never seen in training.

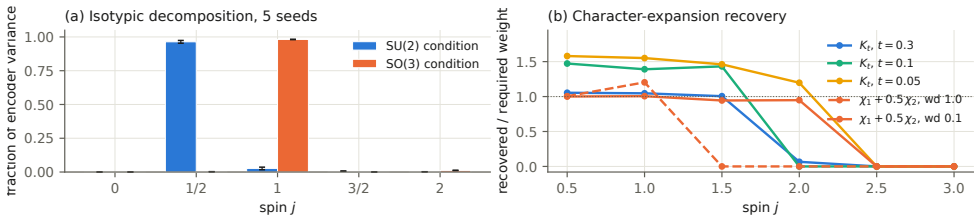


Figure 1: (a) Isotypic decomposition of the trained encoder for the $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ conditions (five seeds). (b) Recovery of the character expansion of the score for heat-kernel and band-limited targets. Values near 1 indicate a component carried at its required weight, and the dashed curve shows the in-support spin-2 component pruned at weight decay 1.0.

Table 1: Learned representations across ten group forms, with the fraction of encoder variance on their matrix coefficients (mean over five seeds) and the rejected faithful alternative where one exists. $\mathrm{Spin}(4)$ representations are labeled by pairs (j_L, j_R). Seed statistics and intervention results per condition are in Appendix D.

task group	learned representation	variance	rejected alternative
$\mathrm{SU}(2)$	spin $\frac{1}{2}$	0.96	
$\mathrm{SO}(3)$	spin 1	0.98	
$\mathrm{SU}(3)$	$\mathbf{3} \oplus \bar{\mathbf{3}}$	0.97	
$\mathrm{PSU}(3)$	adjoint $\mathbf{8}$	0.93	
$\mathrm{Spin}(4)$	$(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$	0.98	(no faithful irreducible)
$\mathrm{SO}(4)$	$(\frac{1}{2}, \frac{1}{2})$	0.97	
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	$(1, 0) \oplus (0, \frac{1}{2})$, 5 dims	0.98	$(1, \frac{1}{2})$, 6 dims: 0.008
$\mathrm{PSO}(4)$	$(1, 0) \oplus (0, 1)$, 6 dims	0.98	$(1, 1)$, 9 dims: 0.011
$\mathrm{Spin}(5)$	spinor $\mathbf{4}$	0.97	
$\mathrm{SO}(5)$	vector $\mathbf{5}$ + adjoint $\mathbf{10}$	$0.69 + 0.22$	(see Section 6)

Encodings. Each element enters through a fixed map Ψ, so the model has no per-element parameters. For $\mathrm{SU}(2)$, we take $\Psi(q)_i = \exp(\alpha\langle q, h_i\rangle)$ with 32 fixed Haar landmarks h_i and $\alpha = 3$, standardized under Haar statistics. The quotient condition for G/Z replaces Ψ by its average over the Z-orbit. The averaged map is exactly Z-invariant and separates the elements of G/Z, so it realizes the quotient group, and the two conditions share landmarks, dimension, and all hyperparameters. Every result below reproduces under two further encodings (Appendix C), so the selected representation is a property of the task and the group.

Model. An encoder-factorized scorer with about 0.3M parameters. A shared two-layer GELU encoder produces $z(a), z(b)$, an interaction network produces $h(a, b) \in \mathbb{R}^{128}$, a candidate encoder produces $z_c(c)$, and the score is $\langle h(a, b), z_c(c)\rangle$, trained by AdamW with weight decay 1.0 unless stated. Every reported quantity is a mean over five seeds with its standard deviation.

Analysis methods. The isotypic decomposition computes, on a fixed Haar probe set, the fraction of variance of the centered activations z lying in the sampled, jointly orthonormalized spans of matrix coefficients, component by component in order of Casimir eigenvalue, with exact coefficients verified in a unit-test suite (Appendix C) and a sampling floor near 0.2% of variance. Ablations subtract from z the part linearly predictable from a chosen component's exact matrix coefficients, an intervention well defined even for components at the floor. Activation patching replaces the internal product state by reconstructions from chosen coordinates. The activations are real, so a complex irreducible representation and its conjugate span one real isotypic component, of dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} \mathrm{End}_G(W)$ (Lemma 2, with background in Appendix A).

3 The learned representation determines the global form

Across the ten group forms of Table 1, the trained features concentrate on the matrix coefficients of a small faithful representation of the task group, the same one in every seed. The half-integer component of $\mathrm{SU}(2)$ appears if and only if the encoding distinguishes $\pm g$, with $\cos(z(g), z(-g)) =$

-0.95 ± 0.02 in the faithful condition and $+1.000$ in the quotient condition. The learned features therefore detect the double cover, a property of the isomorphism type that the Lie algebra does not determine. When the center of the group fails to be cyclic, no faithful irreducible representation exists (Fact 1), and the models build the direct sum that faithfulness forces. Where a smaller reducible faithful sum competes against a larger faithful irreducible representation, the sum wins, with the alternative under 1% of variance in every seed.

The selection is causal. Subtracting from z the part linearly predictable from the learned representation's matrix coefficients collapses ranking to chance (0.02 against a chance rate of 0.016) and collapses center discrimination to its tie value, while the same ablation on any other component, or on a random subspace of equal rank, leaves both intact. The pattern is a double dissociation between each group and its quotient, with seed standard deviations below 0.005 on every cell (Table 3). It repeats under the order-three center of $\mathrm{SU}(3)$, where orbit averaging acts as the identity on quotient models and destroys the faithful ones (Table 4), and it holds summand by summand for $\mathrm{Spin}(4)$, whose center supports three distinct discriminations, with each learned summand necessary for exactly the center elements its central character detects (Table 5). The trained scorer is itself a character function, and a single character explains 0.97 ± 0.01 of the score variance for $\mathrm{SU}(2)$ (with $\chi_{1/2}$) and 0.88 ± 0.01 for $\mathrm{SO}(3)$ (with χ_1).

The following bound, proved in Appendix B, explains why ranking never forces a larger representation.

Lemma 1. *Let V be a faithful finite-dimensional unitary representation of a compact group H with character χ_V. Then $\operatorname{Re}\chi_V(g) \le \chi_V(e)$ for all $g \in H$, with equality only at $g = e$. Moreover, for X in the Lie algebra of H, $\operatorname{Re}\chi_V(\exp X) = \chi_V(e) - \frac{1}{2}\|dV(X)\|_F^2 + O(\|X\|^4)$, and $X \mapsto \|dV(X)\|_F^2$ is positive definite.*

By Lemma 1, a monotone function of $\operatorname{Re}\chi_V(ab\,c^{-1})$ ranks the product above every excluded negative for any faithful V, so optimization selects the representation on cost. Conversely, a representation with nontrivial kernel forces positive ranking error under explicit sampling assumptions (Lemma 3), and the ablations above realize that necessity summand by summand.

4 The learned features span the full coefficient space

The learned features occupy their isotypic component fully. In every seed, the active subspace of z has exactly the real dimension of the full coefficient space, 4 for the quaternionic spinor and 9 for the real-type vector representation, as the following lemma prescribes (proof in Appendix B).

Lemma 2. *Let $\rho\colon G \to U(V_\rho)$ be an irreducible unitary representation of a compact group G, and let $M_\rho \subseteq L^2(G)$ be the span of its matrix coefficients $g \mapsto \langle u, \rho(g)v\rangle$. Then $M_\rho \cong V_\rho^* \boxtimes V_\rho$ as a representation of $G \times G$ acting by left and right translation, this representation is irreducible, and every nonzero subspace of M_ρ invariant under both translation actions equals M_ρ. For an irreducible real representation W with commutant $D = \mathrm{End}_G(W)$, the corresponding real coefficient space has dimension $(\dim_{\mathbb{R}} W)^2/\dim_{\mathbb{R}} D$ and is the minimal nonzero two-sided translation-stable real subspace of its isotypic component.*

Two-sided structure. Lemma 2 makes full-coefficient identity falsifiable. The full coefficient space admits commuting linear left and right translation actions, and every proper subspace of the isotypic component loses at least one of them. We fit affine operators M_k^L, M_k^R with $z(kg) \approx M_k^L z(g)$, $z(gk) \approx M_k^R z(g)$, evaluated held out. On the trained $\mathrm{SO}(3)$ features, both actions fit at $R^2 = 0.989 \pm 0.001$ with relative commutator defect $\|[M_a^L, M_b^R]\| = 0.012 \pm 0.001$, matching the exact coefficient space. On the frame $g \mapsto (ge_1, ge_2)$, the left action is exactly linear and the right action fits at $R^2 = 0.638$, since the missing column $ge_3 = (ge_1) \times (ge_2)$ is uncorrelated with the retained ones under Haar measure. A random subspace of the coefficient space at the frame's dimension loses both actions, fitting at $R^2 = 0.67$ (full comparison in Table 6). The fitted operators realize the representation itself. The trace of M_k^L equals $\dim(W)\,\chi_W(k)$ with correlation 1.000 across group elements, and its eigenvalue phases match those of $\dim(W)$ copies of $W(k)$ to 0.002 radians for $\mathrm{SU}(2)$.

Causal sufficiency. Replacing the internal product state $h(a, b)$ by its ridge reconstruction from the eight realified spinor coefficients of a counterfactual product $a'b'$ redirects the $\mathrm{SU}(2)$ model to select

$a'b'$ with probability 0.996–0.999 in every seed, equal to or better than patching the full activation, while reconstruction from the wrong component redirects at chance. The output h concentrates $89.9 \pm 1.4\%$ of its variance on the diagonal two-argument component of the learned representation, with $98.2 \pm 0.3\%$ of that energy in the subspace spanned by the matrix product $W(a)W(b)$, so the interaction network implements the representation's own composition law.

Cost. To test whether two-sided linearity is the reason the coefficient space wins, we trained matched models on $\mathrm{SO}(3)$ whose encodings are the exact frame with 6 coordinates or the exact coefficient space with 9, all else equal. At statistically equal accuracy (0.891 ± 0.002 against 0.889 ± 0.003), the coefficient-space model attains a larger margin (7.29 ± 0.16 against 6.49 ± 0.05) at 17% lower total parameter norm, with every module cheaper (Table 7). The frame model reconstructs the missing column nonlinearly, and the reconstruction costs more than the three saved coordinates. Regularized training therefore selects the full coefficient space as the cheapest complete mechanism, with storage and composition counted together.

5 Spectral bias in order of the Casimir eigenvalue

Ranking requires a single faithful character, so it leaves the allocation of capacity across isotypic components unobserved. Regression onto the $\mathrm{SU}(2)$ heat kernel $K_t(g) = \sum_j (2j+1) e^{-t\, j(j+1)} \chi_j(g)$, evaluated at $ab\,c^{-1}$, requires every component that the envelope keeps above the noise. Figure 1(b) reports the fraction of score variance per component, sequentially orthogonalized in order of Casimir eigenvalue, divided by the same fraction for the exact target, so a value of 1 means a component at its required weight (per-checkpoint values in Table 8).

Three regularities emerge. First, each trained score is a truncated character expansion, with recovered components near ratio 1, components beyond a sharp truncation point at 0, and the truncation at higher Casimir eigenvalue for smaller t. Second, the truncation advances in order of Casimir eigenvalue during training. At $t = 0.05$, the spin-2 recovery climbs from 0.12 to 1.20 over 70k steps while spin $\frac{5}{2}$ stays at 0. Third, the truncation moves with the regularization strength. On the band-limited target $\chi_1 + \frac{1}{2}\chi_2$, weight decay 1.0 prunes the spin-2 part, the abandoned 12.6% of target variance equals the final test error, and weight decay 0.1 recovers the target with profile $(1.00, 1.01, 0.94, 0.95)$ and out-of-support components at 0.000. A component survives when its variance contribution exceeds its norm cost, so the truncation is a cost boundary of the same kind that selects the representation in Section 3. The central character of the target governs the parity of the learned features as well. The band-limited target is invariant under the center, and the model builds an integer-spin representation even on the faithful $\mathrm{SU}(2)$ encoding, with the spinor component at 1%.

6 A negative finding, and a selection conjecture

On $\mathrm{SO}(5)$, the learned features occupy a mixture, the vector $\mathbf{5}$ at 0.68–0.70 of variance and the adjoint $\mathbf{10}$ at 0.21–0.23, stable across five seeds, doubled training, four times the candidates, and reduced weight decay. The adjoint fraction falls only when generalization improves, from 0.23 at held-out accuracy 0.55 to 0.17 at 0.73 under tripled data, and no condition solves the task. The minimal-representation description therefore fails as stated in one of the ten group forms. Both faithful components ranking correctly is consistent with Lemma 1, and two hypotheses remain open, purification as the task approaches a solution, or a genuinely mixed minimum-cost mechanism for this group, which the cost result of Section 4 makes coherent. Separating them requires a condition that solves $\mathrm{SO}(5)$.

The structural results suggest a selection principle that we can state but cannot yet prove.

Conjecture 1. *In the factorized architecture with weight decay, among parameter settings achieving population ranking loss within ε of the optimum, the minimum-norm setting realizes, as $\varepsilon \to 0$, the full real coefficient space of a faithful representation minimizing the total regularized cost of features, composition, and scoring, with score a monotone function of* $\mathrm{Re}\,\chi_V(ab\,c^{-1})$.

The missing ingredients are a norm lower bound tying any ε-optimal score to the effective dimension of the functions it computes on G, and the exclusion of solutions outside the character algebra, the compact-group analogue of the alternative circuits documented for finite groups Stander et al. [2024],

Zhong et al. [2023]. Every quotient condition in our experiments generalizes below its cover, and disentangling representation selection from the conditioning of the quotient encoding is left as future work.

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A Background: representation theory and harmonic analysis on compact groups

This appendix collects the facts the paper uses, with references in place of proofs. Standard sources are Bröcker and tom Dieck [1985] for compact groups and the Peter–Weyl theorem, Sepanski [2007] for the real Schur lemma and the type classification, Folland [1995] for harmonic analysis, and Fulton and Harris [1991] for the representation theory of the classical groups. Throughout, G is a compact Lie group and all representations are finite-dimensional and continuous.

Haar measure and unitarity. A compact group carries a unique translation-invariant Borel probability measure, its Haar measure, and every expectation over group elements in this paper is taken with respect to it. Averaging an arbitrary inner product over the group makes any representation unitary, and every unitary representation decomposes as a finite orthogonal direct sum of irreducible ones [Bröcker and tom Dieck, 1985, Ch. II]. The set of isomorphism classes of complex irreducible representations is denoted $\widehat{G}$, and for $\rho \in \widehat{G}$ we write $d_\rho = \dim V_\rho$ and $\chi_\rho(g) = \operatorname{tr} \rho(g)$ for its character.

Matrix coefficients and orthogonality. For $\rho \in \widehat{G}$ and vectors $u, v \in V_\rho$, the function $g \mapsto \langle u, \rho(g)v \rangle$ is a matrix coefficient of ρ. Schur's lemma states that a linear map commuting with two irreducible representations is zero or an isomorphism, and that the self-intertwiners of a complex irreducible representation are the scalars [Fulton and Harris, 1991, Ch. I]. Its integrated form is the orthogonality of coefficients. In a unitary basis,

$$\int_G \rho(g)_{ij}\, \overline{\sigma(g)_{kl}}\, dg \;=\; \frac{\delta_{\rho\sigma}\, \delta_{ik}\, \delta_{jl}}{d_\rho}, \qquad \int_G \chi_\rho(g)\, \overline{\chi_\sigma(g)}\, dg \;=\; \delta_{\rho\sigma}$$

for $\rho, \sigma \in \widehat{G}$ [Bröcker and tom Dieck, 1985, Ch. II]. These relations underlie the isotypic decomposition of Section 2. They make the coefficient spans of distinct irreducible representations orthogonal in $L^2(G)$, so variance fractions per component are well defined, and our implementation verifies them numerically on the sampled probe sets.

The Peter–Weyl theorem. The matrix coefficients of all $\rho \in \widehat{G}$ span a dense subspace of $L^2(G)$, and as a representation of $G \times G$ acting by left and right translation,

$$L^2(G) \;\cong\; \bigoplus_{\rho \in \widehat{G}} M_\rho, \qquad M_\rho \;\cong\; V_\rho^* \boxtimes V_\rho,$$

an orthogonal Hilbert-space direct sum in which each coefficient space M_ρ has dimension d_ρ^2 [Bröcker and tom Dieck, 1985, Ch. III]. Every function a network computes on G therefore has a spectral decomposition over $\widehat{G}$, and the structural question of this paper, which components a trained model occupies and whether it occupies them fully, is a question about this decomposition. Lemma 2 records the two-sided irreducibility of M_ρ and its real form.

Real representations and Schur type. The networks compute with real numbers, so the relevant decomposition is that of real representations. For an irreducible real representation W, the real Schur lemma states that the commutant $\operatorname{End}_G(W)$ is a real division algebra, hence isomorphic to $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$ [Sepanski, 2007, Ch. III], and the three cases correspond to the classical trichotomy for a complex irreducible representation V. The representation V has real type when it admits an invariant real structure, quaternionic type when it admits an invariant quaternionic structure, and complex type when $V \not\cong \bar{V}$. Realification acts on dimensions as follows. A real-type V of dimension d yields $\dim_{\mathbb{R}} W = d$ with commutant $\mathbb{R}$. A complex-type or quaternionic-type V of dimension d yields $\dim_{\mathbb{R}} W = 2d$ with commutant $\mathbb{C}$ or $\mathbb{H}$, and in the complex case V and $\bar{V}$ realify to the same W, which is why a complex irreducible representation and its conjugate span a single real isotypic component in our measurements. In all three cases, the real coefficient space has dimension

$$\dim_{\mathbb{R}} M_W \;=\; \frac{(\dim_{\mathbb{R}} W)^2}{\dim_{\mathbb{R}} \operatorname{End}_G(W)},$$

which evaluates to d^2, $2d^2$, and d^2 in the real, complex, and quaternionic cases. The three dimensions measured in the paper are instances, namely 9 for the real-type vector representation of $\mathrm{SO}(3)$, 18 for the complex-type fundamental pair of $\mathrm{SU}(3)$, and 4 for the quaternionic spinor of $\mathrm{SU}(2)$.

Characters, class functions, and the scoring identity. Characters are conjugation-invariant, and the irreducible characters form an orthonormal basis of the class functions in $L^2(G)$ [Bröcker and tom Dieck, 1985, Ch. II]. Two identities are used throughout, namely $\chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$ and, for unitary ρ,

$$\chi_\rho(ab\,c^{-1}) \;=\; \mathrm{tr}\big(\rho(a)\rho(b)\rho(c)^*\big) \;=\; \big\langle \rho(c),\, \rho(a)\rho(b)\big\rangle_{\mathrm{HS}},$$

so a character evaluated at $ab\,c^{-1}$ is a Hilbert–Schmidt pairing of the candidate's coefficients against the product of the arguments' coefficients. This identity supports the statement in Section 4 that composition and scoring become bilinear on the coefficient space, and it is the form in which the finite-group mechanism of Chughtai et al. [2023] extends to the compact case.

Central characters and descent to quotients. By Schur's lemma, the center $Z(G)$ acts in any $\rho \in \widehat{G}$ by a character $\omega_\rho \colon Z(G) \to U(1)$, called the central character. For a closed central subgroup Z, the irreducible representations of the quotient G/Z are exactly those ρ with ω_ρ trivial on Z, and the spectral support of a Z-invariant function on G consists of such ρ alone. This is the descent rule of Section 3. The orbit-averaged encoding is Z-invariant, so every component carrying a nontrivial central character of Z is forbidden, and the measured floors on those components are the empirical form of the rule. One consequence used for $\mathrm{Spin}(4)$ deserves a proof, since it forces the direct-sum representations of Table 1.

Fact 1. *If a compact group admits a faithful irreducible complex representation, its center is isomorphic to a closed subgroup of $U(1)$. In particular, $\mathrm{Spin}(4)$, with center $\mathbb{Z}_2 \times \mathbb{Z}_2$, has no faithful irreducible representation.*

Proof. Under a faithful irreducible ρ, the center acts by the scalars ω_ρ, and faithfulness makes $z \mapsto \omega_\rho(z)$ injective, so $Z(G)$ embeds in $U(1)$. Every finite subgroup of $U(1)$ is cyclic, and $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not. □

The Casimir element and the heat kernel. Fix a bi-invariant metric on G. The quadratic Casimir element of the universal enveloping algebra acts in each irreducible representation ρ as a scalar $C(\rho) \geq 0$, and the matrix coefficients of ρ are eigenfunctions of the Laplace–Beltrami operator on G with eigenvalue $C(\rho)$ [Sepanski, 2007, Ch. 5]. The Casimir eigenvalue therefore plays the role of a squared frequency, and ordering components by Casimir eigenvalue in this paper is the compact-group analogue of ordering Fourier modes by frequency. The heat kernel at the identity-based variable g is the class function

$$K_t(g) \;=\; \sum_{\rho \in \widehat{G}} d_\rho\, e^{-t\,C(\rho)}\, \chi_\rho(g),$$

the fundamental solution of the heat equation on G [Folland, 1995, Ch. 1], and its envelope $e^{-t\,C(\rho)}$ determines the required component weights in Section 5.

The groups of the experiments. The irreducible representations of $\mathrm{SU}(2)$ are the spin representations D^j of dimension $2j+1$ for $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, with Casimir eigenvalue $C(D^j) = j(j+1)$, quaternionic type for half-integer j and real type for integer j, and central character $(-1)^{2j}$ at $-1 \in \mathrm{SU}(2)$ [Bröcker and tom Dieck, 1985, Ch. II]. The quotient $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm 1\}$ therefore carries exactly the integer spins, which yields the double-cover diagnostic of Section 3. The irreducible representations of $\mathrm{SU}(3)$ are labeled by highest weights (p, q) with dimension $\frac{1}{2}(p+1)(q+1)(p+q+2)$ and central character ω^{p-q} at the central element ωI for ω a primitive cube root of unity, so descent to $\mathrm{PSU}(3)$ requires $p \equiv q \pmod 3$. The components tracked in the experiments are the fundamental pair $\mathbf{3} \oplus \bar{\mathbf{3}}$ at $(1,0)$ and $(0,1)$, the adjoint $\mathbf{8}$ at $(1,1)$, and the pairs $\mathbf{6} \oplus \bar{\mathbf{6}}$ and $\mathbf{10} \oplus \bar{\mathbf{10}}$ [Fulton and Harris, 1991, Ch. 13]. The group $\mathrm{Spin}(4)$ is isomorphic to $\mathrm{SU}(2) \times \mathrm{SU}(2)$, its irreducible representations are the outer products (j_L, j_R) with central character $\big((-1)^{2j_L}, (-1)^{2j_R}\big)$ on the center $\mathbb{Z}_2 \times \mathbb{Z}_2$, and its three central quotients are $\mathrm{SO}(4)$ from the diagonal $\mathbb{Z}_2$, retaining $j_L + j_R \in \mathbb{Z}$, the product $\mathrm{SO}(3) \times \mathrm{SU}(2)$ from one factor's $\mathbb{Z}_2$, and $\mathrm{PSO}(4)$ from the full center, retaining $j_L, j_R \in \mathbb{Z}$. The group $\mathrm{Spin}(5)$ is isomorphic to the quaternionic unitary group $\mathrm{Sp}(2)$, with quaternionic spinor representation $\mathbf{4}$, real vector representation $\mathbf{5}$, and adjoint $\mathbf{10}$, and $\mathrm{SO}(5) = \mathrm{Spin}(5)/\{\pm 1\}$ retains the representations with trivial central character, among them $\mathbf{5}$ and $\mathbf{10}$ [Fulton and Harris, 1991, Ch. 16].

B Proofs

Proof of Lemma 1. Each eigenvalue of the unitary matrix $V(g)$ lies on the unit circle, so $\operatorname{Re}\operatorname{tr} V(g) \le \dim V$, with equality exactly when every eigenvalue equals 1, that is, when $V(g)$ is the identity. By faithfulness, this occurs only at $g = e$. For the expansion, write $A = dV(X)$, an anti-Hermitian matrix with eigenvalues $i\lambda_1, \dots, i\lambda_{\dim V}$. Then $\operatorname{Re}\operatorname{tr}\exp(tA) = \sum_k \cos(t\lambda_k) = \dim V - \frac{t^2}{2}\sum_k \lambda_k^2 + O(t^4)$, and $\sum_k \lambda_k^2 = \|A\|_F^2$. The trace of A is purely imaginary and contributes nothing to the real part. If $\|dV(X)\|_F = 0$, then the one-parameter subgroup $\exp(tX)$ lies in the kernel of V, which is trivial, so $X = 0$, proving positive definiteness. □

Lemma 3. *Let H be a compact group. Assume the score factors as $s(a, b, c) = S(\varphi(a), \varphi(b), \varphi(c))$ where φ factors through the matrix coefficients of a representation V of H with nontrivial closed kernel N, the function $c \mapsto s(a, b, c)$ is continuous, negatives are sampled from a distribution assigning positive probability to every open subset of the admissible region, and the exclusion radius is smaller than $\min_{n \in N,\, n \neq e} d(e, n)$. Then the population ranking loss is bounded away from its optimum.*

Proof. Fix $n \in N$ with $n \neq e$. By the radius hypothesis, $ab\,n$ lies in the admissible negative region. Since φ is constant on N-cosets, $s(a, b, ab\,n) = s(a, b, ab)$, which is at least the score of the noisy positive up to the continuity modulus of s over the positive displacement. By continuity, there is a ball around $ab\,n$ on which the score stays within any prescribed gap of this value, and by the sampling hypothesis that ball receives positive probability, so with positive probability a sampled negative ties or exceeds the positive. □

The hypotheses of Lemma 3 are needed. With a finite kernel and a continuous candidate distribution, the exact ambiguous points have measure zero, and without continuity of the score no positive lower bound follows from non-separation alone.

Proof of Lemma 2. The map $V_\rho^* \otimes V_\rho \to L^2(G)$ sending $u^* \otimes v$ to $g \mapsto \langle u, \rho(g)v\rangle$ intertwines the $G \times G$-action by left and right translation with the outer tensor product action and is injective by the orthogonality of matrix coefficients (Appendix A). The outer tensor product of irreducible representations is an irreducible representation of the product group, so M_ρ has no proper nonzero two-sided invariant subspace, and $\dim_{\mathbb{C}} M_\rho = (\dim V_\rho)^2$. For the real form, the real Schur lemma gives $\operatorname{End}_G(W) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and the real matrix-coefficient span of W is identified with $W^* \otimes_D W$, of real dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} D$. Minimality follows from irreducibility of the two-sided action on this space, by the same outer-tensor argument applied over D. □

C Experimental details

Groups and sampling. Elements of $\mathrm{SU}(2)$ are unit quaternions, of $\mathrm{SU}(3)$ Haar-random special unitary matrices, of $\mathrm{Spin}(4) = \mathrm{SU}(2) \times \mathrm{SU}(2)$ pairs of unit quaternions, and of $\mathrm{Spin}(5) = \mathrm{Sp}(2)$ quaternionic 2×2 unitaries sampled by quaternionic Gram–Schmidt. Matrix coefficients for half-integer spins of $\mathrm{SU}(2)$ are computed exactly through the action on homogeneous polynomials, for $\mathrm{SU}(3)$ through symmetric powers of the fundamental representation and the adjoint action on the Gell-Mann frame, and for $\mathrm{Sp}(2)$ through the complex embedding of the quaternionic matrices together with the conjugation actions on Hermitian and anti-Hermitian quaternion matrices.

Quotient encodings. Quotient conditions average the encoding over the central subgroup being quotiented, namely the antipodal average for $\mathrm{SO}(3)$ and $\mathrm{SO}(5)$, the order-three orbit average for $\mathrm{PSU}(3)$, and the three sign-pair averages for the quotients of $\mathrm{Spin}(4)$. Each average is exactly invariant under the quotiented subgroup and distinguishes all other center elements, which we verify numerically at construction. The two further encodings under which the headline results reproduce are a frozen random two-layer tanh network of the raw coordinates and the values of a fixed random Fourier function at 32 translated probe points, each with its orbit-averaged quotient variant.

Training. Batch size 4096 (2048 for the larger groups), learning rate 10^{-3}, weight decay 1.0, 100k–200k steps, training fraction 0.01 of the pair table unless stated. The ranking results of Table 1 use five seeds per condition. Heat-kernel and band-limited regressions replace the cross-entropy by mean squared error on standardized targets, with candidates drawn half near the product at uniform

Table 2: Seed statistics for the principal conditions (five seeds, mean $\pm$ standard deviation, evaluated at 100k steps). D denotes the center discrimination appropriate to the condition, the probability that the true product outscores its center translates, with tie value 0.5 ($\mathrm{SU}(2)$/$\mathrm{SO}(3)$) or $1/3$ ($\mathrm{SU}(3)$/$\mathrm{PSU}(3)$).

	$\mathrm{SU}(2)$	$\mathrm{SO}(3)$	$\mathrm{SU}(3)$	$\mathrm{PSU}(3)$
held-out accuracy	0.921 ± 0.003	0.883 ± 0.003	0.994 ± 0.001	0.685 ± 0.020
learned-representation variance	0.835 ± 0.010	0.917 ± 0.008	0.971 ± 0.009	0.934 ± 0.007
forbidden-component variance	0.0023 ± 0.0004	0.0023 ± 0.0004	0.0059 ± 0.0004	0.0000
center cosine	-0.72 ± 0.02	$+1.000$	-0.484 ± 0.005	$+1.000$
coefficient decode R^2	0.995	0.999	0.978 ± 0.006	0.699 ± 0.012
D, intact	1.0000	0.5000	1.0000	0.3333
score–character R^2	0.974 ± 0.006	0.876 ± 0.014	0.700	

Table 3: Interventions on the argument encoders for the $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ conditions (five seeds, mean $\pm$ standard deviation). Task is ranking accuracy against 63 excluded Haar negatives (chance 0.016), D_c the probability that the true product outscores its antipode (tie 0.5).

	$\mathrm{SU}(2)$		$\mathrm{SO}(3)$	
intervention	task	D_c	task	D_c
none	1.000	1.0000 ± 0.0000	1.000	0.5000
ablate spin $\frac{1}{2}$	0.026 ± 0.002	0.503 ± 0.006	1.000 ± 0.000	0.5000
ablate spin 1	1.000 ± 0.000	1.0000	0.027 ± 0.003	0.5000
center-orbit average	0.021	0.498	1.000 (exact identity)	0.5000
random subspace, equal rank	1.000	1.0000	1.000	0.5000

geodesic radius and half from Haar measure. Under the $\mathrm{SU}(2)$ sampling, the Bayes-optimal ranking rule attains accuracy 0.951.

Analyses. Isotypic variance fractions use jointly orthonormalized sampled coefficient spans processed in order of Casimir eigenvalue, which bounds cross-component leakage at the sampling floor. Ablations subtract the part of z linearly predictable from a component's exact matrix coefficients, fitted on an independent probe set. Subspace estimates of near-floor components are noise-dominated and ablate incorrectly, which is why the function-space form is the reported intervention. Ridge probes select the regularization strength on a validation split and report on a disjoint test split. Translation operators are fitted by ridge regression in a PCA frame retaining 99.9% of activation variance, with homomorphism and commutator defects evaluated on held-out elements. The recovery profiles regress both the learned score and the exact target on the same sequentially orthogonalized character design, so standardization and collinearity cancel in the reported ratios. Raw per-character least-squares coefficients are unidentifiable under this sampling because high-Casimir characters are near-collinear near the identity, and the sequentially orthogonalized profile is the identifiable replacement.

Verification. The representation-theoretic primitives pass a unit-test suite covering the homomorphism property, unitarity, characters, central characters, Schur orthogonality, planted-component recovery, and exact invariance of the quotient encodings. The recovery analysis reproduces the known support and coefficients of the band-limited target at weight decay 0.1, which calibrates the method. The orbit-average intervention acts as the identity on quotient models by construction and measures as one, which confirms the intervention machinery. Software versions are PyTorch 2.11 and NumPy on a single T4 GPU. A tracked results bundle records, for every run, the configuration, seed, checkpoint hash, final metrics, and analysis outputs, and every table in this paper regenerates from the bundle. Code and the bundle are included as anonymized supplementary material.

D Complete results

Figure 2 shows the training dynamics of the isotypic decomposition and the per-group component fractions behind Table 1. Table 2 lists the seed statistics for the four principal conditions. Tables 3 to 5 list the intervention results, Table 6 the two-sided translation test, Table 7 the frame comparison, and Table 8 the recovery of the character expansion per checkpoint.

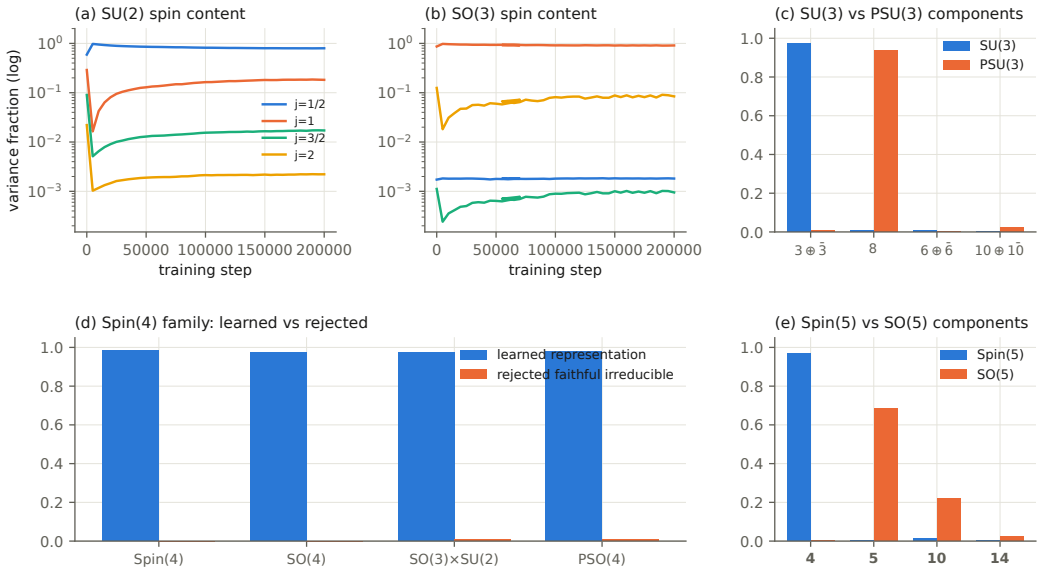


Figure 2: (a, b) Fraction of encoder variance per spin component during training for the $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ conditions (seed 0, log scale, with the dotted floor marking the sampling limit). (c, d, e) Final component fractions for $\mathrm{SU}(3)$/$\mathrm{PSU}(3)$, the $\mathrm{Spin}(4)$ family, and $\mathrm{Spin}(5)$/$\mathrm{SO}(5)$ (means over five seeds).

Table 4: Interventions for the $\mathrm{SU}(3)$ and $\mathrm{PSU}(3)$ conditions (five seeds). D_ω is the probability that the true product outscores both of its center translates (tie $1/3$).

	SU(3)		PSU(3)	
intervention	task	D_ω	task	D_ω
none	0.998 ± 0.001	1.000	0.739 ± 0.019	0.333
center-orbit average	0.016 ± 0.001	0.337 ± 0.008	0.739 (exact identity)	0.333
remove invariant part	0.998	1.000	0.015 ± 0.001	0.333
ablate $\mathbf{3} \oplus \bar{\mathbf{3}}$	0.018 ± 0.002	0.335	0.733	0.333
ablate adjoint $\mathbf{8}$	0.997	1.000	0.014 ± 0.003	0.333

Table 5: Interventions resolved by summand for the $\mathrm{Spin}(4)$ family (five seeds, representative cells with seed statistics). D_L, D_R, D_{LR} are the discriminations of the three center elements (tie 0.5). Each learned summand is necessary for exactly the center elements its central character detects.

condition	intervention	task	D_L	D_R
$\mathrm{Spin}(4)$	none	0.997	1.000	1.000
$\mathrm{Spin}(4)$	ablate $(\frac{1}{2}, 0)$	0.57	0.507 ± 0.007	1.000 ± 0.000
$\mathrm{Spin}(4)$	ablate $(0, \frac{1}{2})$	0.57	1.000	0.498
$\mathrm{SO}(4)$	ablate $(\frac{1}{2}, \frac{1}{2})$	0.017 ± 0.001	0.489	0.489
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	ablate $(0, \frac{1}{2})$	0.53	0.500	0.493 ± 0.006
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	ablate $(1, 0)$	0.853 ± 0.036	0.500	1.000
$\mathrm{PSO}(4)$	ablate $(1, 0)$ or $(0, 1)$	0.809 ± 0.016	0.500	0.500

Table 6: The two-sided translation test on the $\mathrm{SO}(3)$ condition (five seeds for the trained and random-initialization systems). Left and right columns are held-out R^2 of fitted affine translation operators, hom-R the right-homomorphism defect, and $[L, R]$ the commutator defect, both relative to activation norm.

feature system	dim	left R^2	right R^2	hom-R	$[L, R]$
trained features	9	0.989 ± 0.001	0.989 ± 0.001	0.012	0.012 ± 0.001
full coefficient space	9	1.000	1.000	0.000	0.000
two-column frame	6	1.000	0.638	0.526	0.039
random subspace of coefficients	6	0.674	0.675	0.411	0.448
raw encoding	32	0.992	0.991	0.058	0.073
random-initialization encoder	16	0.892 ± 0.008	0.890 ± 0.008	0.179	0.160 ± 0.014

Table 7: Frame against coefficient space in matched training on $\mathrm{SO}(3)$ (five seeds per arm, 60k steps). Norms are Frobenius norms of module parameters at the end of training.

encoding	accuracy	margin	encoder norm	interaction norm	scorer norm	margin/norm
frame, 6 dims	0.891 ± 0.002	6.49 ± 0.05	19.6	11.9	13.7	0.243 ± 0.002
coefficients, 9 dims	0.889 ± 0.003	7.29 ± 0.16	16.7	9.8	12.2	0.319 ± 0.008

Table 8: Recovery of the character expansion of the score per checkpoint (sequentially orthogonalized profile, model divided by target, with components beyond spin 2 at 0.000 throughout and omitted). The band-limited target is $\chi_1 + \frac{1}{2}\chi_2$.

target	step	$j = \frac{1}{2}$	$j = 1$	$j = \frac{3}{2}$	$j = 2$
$K_t, t = 0.3$	25k	1.13	1.12	0.13	0.81
$K_t, t = 0.3$	60k	1.05	1.05	1.01	0.07
$K_t, t = 0.1$	60k	1.47	1.39	1.43	0.00
$K_t, t = 0.05$	50k	1.81	1.81	1.71	0.12
$K_t, t = 0.05$	100k	1.63	1.63	1.49	0.83
$K_t, t = 0.05$	120k	1.58	1.55	1.46	1.20
band-limited, wd 1.0	60k	1.01	1.20	0.00	0.00
band-limited, wd 0.1	60k	1.00	1.01	0.94	0.95